

STATEMENT OF WORK

1. Objective/Requirements

This requirement is for the acquisition of FY25 CIVA Maintenance for NASA Johnson Space Center (JSC) for use in x-ray detection of cracks and other flaws.

2. Characteristics, Scope, and Specs

- The Chosen Contractor shall authorize NASA Johnson Space Center to use the CIVA Software on a PC connected to a local network license (RT module). With a network license, the remote desktop mode is available.
- NASA Johnson Space Center shall specify the name of the site or department that will be using the CIVA Software, along with a technical employee "correspondent" to interface with EXTENDE regarding any technical issues or errors related to the use of the CIVA Software.

QTY	DESCRIPTION
1	Extende Additional Year of Maintenance (for CIVA Software RT Module)
1	CIVA 2023 RT Module Permanent Local Network License

Support Services shall include and are limited to the following:

- Access for one (1) year to all updates and new versions of CIVA Software corresponding to the modules in the NASA Johnson Space Center license (RT module) that are included in the maintenance fee set forth herein. This proposal includes an option to purchase an additional year of support services for a total of (2) years if the option is purchased.
- Technical support by e-mail (supportciva@extende.com) in the event of malfunctions, whereby EXTENDE agrees to provide appropriate responses to user messages at latest 10 working days after they are received. For your information, 80% of support questions are answered within 2 days. Unless otherwise agreed by both parties in writing, technical support will be operated by the EXTENDE support team including a European team. EXTENDE shall exercise commercially reasonable efforts to correct any error reported by NASA Johnson Space Center.

Support services and upgrades associated with this CIVA license are provided by the chosen Contractor for one year for free. After this one-year period, the chosen Contractor shall provide support services and upgrades for a charge pursuant to a maintenance contract, to be entered into by the parties.

General Specifications:

- 64 bits OS == Windows 10 or Windows 11
- Graphic Resolution == 1920 x 1080
- No 4k screens
- CIVA shall be installed at the root of a disk $\geq 256\text{Gb}$ if System disk, $\geq 128\text{Gb}$ if dedicated disk
- Read/Write rights are necessary in the CIVA directory (but the cache can be moved => contact us in this case)

Minimal configuration:

- ≥ 4 physical cores (I5 new generation type)
- Ram $\geq 16\text{Gb}$
- SSD Hard Drive (mainly for UT Analysis but all modules use disk cache => SSD has an important impact on performances)
 - Non adapted to launch parametric or POD studies and for modules AUT-Pipeline, CT, DS, SG-Tube, GWT and SHM
 - All modules can run with this type of hardware but with limited configurations and with potentially really low performances

Comfortable configuration**:

- 8 physical cores (i7/i9 types)
- Ram $\geq 32\text{Gb}$
- Dedicated graphic card (also for laptop)
- ** Minimal configuration to smoothly run parametric or POD studies, as well as the modules AUT-Pipeline, CT, DS, SG-Tube, GWT and SHM
- ** Most of simulations and processing will be compatible with this configuration. Several “extreme” use cases and modules could be more demanding.

Optimized configuration***:

- > 8 physical cores
- Ram $\geq 64\text{Gb}$ (128/256 Gb can be necessary for several specific configurations and codes)
- SSD Hard Drive $\geq 500\text{Gb}$
- Dedicated graphic card (also for laptop)
 - *** An efficient use of the CIVA CT module needs this configuration (or at least the comfortable one) and can also require GPU processor for CT reconstructions (GPU is only useful in CT), « CUDA » type with computations capacities: $3.5 \leq \text{version} \leq 9.0$
http://en.wikipedia.org/wiki/CUDA#Supported_GPUs
 - b) Technical support is provided in accordance with the clauses of the Terms below, and is generally available to trained users to address any errors or malfunctions.

- c) CIVA is the state of the art regarding modeling of NDT, EXTENDE can not be held responsible for discrepancies with the experimental results which are due to model inaccuracies or malfunctions of CIVA.
- d) EXTENDE Inc. is dedicated to CIVA Software user service. Upon request, EXTENDE Inc. will provide quotations for additional services related to use of the CIVA Software, for example as follows:
 - Training: general training or dedicated to particular subjects;
 - Definition, creation of configurations for complex calculations cases;
 - Series of calculations (with parametric variations for example);
 - Evaluation of inspection or analysis of procedures by simulation;
 - Or any NDE expert counseling.

3. Place of Performance

E-delivery to NASA Johnson Space Center (JSC)

4. Period of Performance

20 days ARO